Kinds of combinatorial problem

Arrangements, permutations and selections — and the one question that tells them apart.

LESSON 158–160 | 3 × 40 MIN | MATEMATIKA 7, §66 | S8 13

LESSON CLOCK

15' | 30' | 30' | 30' | 15'

The question that decides everything	15 min
Permutations and the factorial	30 min
Arrangements of k from n	30 min
Selections	30 min
Telling them apart	15 min

LEARNING OBJECTIVES

- Ask whether order matters, and answer accordingly.
- Count permutations of n objects with $n!$.
- Count arrangements of k objects chosen from n.
- Count selections where order does not matter.

TEXTBOOK

National	pp. 473–483
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.2

01 Explanation

The question that decides everything

Before counting anything, ask: **does the order matter?** Two problems with identical numbers can have different answers because of it.

PROBLEM	ORDER?	BECAUSE
first and second prize among 5 pupils	yes	first is not second
two pupils on duty from 5	no	the pair is the same either way
a 4-figure PIN	yes	$1234 \neq 4321$
a hand of 4 cards	no	the hand is a set

ORDER DOUBLES OR HALVES THE ANSWER

From 5 pupils, ordered pairs number 20 and unordered pairs 10. Every ordered pair appears twice in the ordered count — once each way round.

Permutations and the factorial

A **permutation** arranges all n objects in a row. There are n choices for the first place, $n - 1$ for the second, and so on.

$$n! = n \cdot (n - 1) \cdot \ldots \cdot 2 \cdot 1$$

N	$N!$	MEANING
1	1	one object, one order
2	2	AB, BA
3	6	six orders of three letters
4	24	seating four pupils
5	120	five books on a shelf
6	720	growing fast

THE FACTORIAL GROWS FASTER THAN ANYTHING MET SO FAR

$10! = 3\,628\,800$. Ten books can be shelved in more than three and a half million orders — a good reason never to try listing them.

Arrangements of k from n

When only k of the n objects are used and the order matters, the product simply stops after k factors.

$$n \cdot (n - 1) \cdot \ldots \text{ — } k \text{ factors}$$

PROBLEM	WORKING	ANSWER
first and second prize among 5	$5 \cdot 4$	20
gold, silver and bronze among 8	$8 \cdot 7 \cdot 6$	336
3 of 10 digits, all different	$10 \cdot 9 \cdot 8$	720
president and secretary from 12	$12 \cdot 11$	132

A PERMUTATION IS THE CASE $K = N$

Arranging 5 of 5 gives $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5!$. The two ideas are one idea, stopped at different points.

Selections

When the order does **not** matter, count the ordered arrangements first and then divide by the number of orders of the chosen group.

$$\text{selections} = \frac{\text{arrangements}}{\text{k!}}$$

PROBLEM	ORDERED	DIVIDE BY	ANSWER
2 pupils on duty from 5	$5 \cdot 4 = 20$	$2! = 2$	10
3 pupils from 6	$6 \cdot 5 \cdot 4 = 120$	$3! = 6$	20
handshakes among 10 people	$10 \cdot 9 = 90$	2	45
diagonals and sides of a hexagon	$6 \cdot 5 = 30$	2	15

HANDSHAKES, DIAGONALS AND PAIRS ARE ONE PROBLEM

Every pair of people shakes once, every pair of vertices joins once. The hexagon's 15 pairs are its 6 sides and 9 diagonals — a check worth making.

Telling them apart

WORDING	ORDER?	KIND
“in how many orders”	yes	permutation
“first, second and third”	yes	arrangement
“a code” or “a number”	yes	arrangement
“choose a group of”	no	selection
“how many pairs”	no	selection
“how many handshakes”	no	selection

PRIZES ARE ORDERED; DUTIES USUALLY ARE NOT

“First and second prize” distinguishes the two places; “two pupils on duty” does not. Reading the wording carefully is the whole of the skill — the arithmetic afterwards is easy.

02 Worked examples

EXAMPLE 1 **In how many ways can 5 books be arranged on a shelf?**

1. All five are used and the order matters.
2. $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$
3. $= 120$

ANSWER

120

EXAMPLE 2 **From *8* runners, in how many ways can gold, silver and bronze be awarded?**

1. Three of eight, and the order matters.
2. $8 \cdot 7 \cdot 6$
3. $= 336$

ANSWER

336

EXAMPLE 3 **How many handshakes take place if *10* people each shake hands once with every other?**

1. Ordered pairs: $10 \cdot 9 = 90$.
2. Each handshake was counted twice.
3. $90 \div 2 = 45$

ANSWER

45

03 Interactive model

Have five pupils stand at the front and ask first for prize orders, then for duty pairs; the same five children give 20 and 10, and the distinction is made physically.

Does the order matter?

First and second prize among 5:

- order matters
- order does not
- neither
- both

So the count is:

- 10
- 20
- 25
- 120

Two pupils on duty from 5:

- order matters
- order does not
- neither
- both

So the count is:

- 10
- 20
- 25
- 120

$5!$ equals:

- 25
- 60
- 120
- 720

Gold, silver and bronze among 8:

- 24
- 56
- 336
- 512

Handshakes among 10 people:

- 45
- 90
- 100
- 20

Choosing 3 pupils from 6:

- 18
- 20
- 120
- 720

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Arrangement	O‘rinlashtirish	Размещение
Permutation	O‘rin almashtirish	Перестановка
Selection	Tanlash	Сочетание
Factorial	Faktorial	Факториал
Order matters	Tartib muhim	Порядок важен
Order does not matter	Tartib muhim emas	Порядок не важен
Committee	Guruh	Комиссия
Handshake	Qo‘l berish	Рукопожатие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A permutation arranges:

- some of the objects
- all of the objects
- two objects
- none

$4!$ equals:

- 12
- 16
- 24
- 64

Order matters in:

- a committee
- a PIN
- a handshake
- a pair

Order does not matter in:

- a race result
- a code
- a group on duty
- a queue

To go from arrangements to selections you:

- multiply by $k!$
- divide by $k!$
- add k
- do nothing

Handshakes among n people number:

- n^2
- $\frac{n(n-1)}{2}$
- $n(n-1)$
- $2n$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3!$

ANSWER 6

2. $4!$

ANSWER 24

3. $5!$

ANSWER 120

4. $6!$

ANSWER 720

5. Arranging 5 books

ANSWER 120

6. First and second prize among 5

ANSWER 20

7. Two on duty from 5

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Gold, silver, bronze among *8*

   ANSWER *336*

2. President and secretary from *12*

   ANSWER *132*

3. Choosing *3* from *6*

   ANSWER *20*

4. Handshakes among *10*

   ANSWER *45*

5. Pairs of vertices of a hexagon

   ANSWER *15*

6. Diagonals of a hexagon

   ANSWER *9*

7. Seating *6* pupils in *6* chairs

   ANSWER *720*

Hard · stretch and reason

7 PROBLEMS

1. Handshakes among n people

   ANSWER $\frac{n(n-1)}{2}$

2. Diagonals of an n-gon

   ANSWER $\frac{n(n-3)}{2}$

3. Choosing *2* from *20*

   ANSWER *190*

4. Arranging *4* of *7* books on a shelf

   ANSWER *840*

5. How many *3*-letter “words” from *АБВГД* with no repeats?

   ANSWER *60*

6. Why divide by $k!$ for a selection?

   ANSWER Each group has been counted in all its orders

7. *10!* is about

   ANSWER *3.6* million

07 Homework

Homework — 5 tasks

Write “order matters” or “order does not matter” beside every question before counting.

1. In how many ways can *6* books be arranged on a shelf?
2. From *9* athletes, in how many ways can the first three places be awarded?
3. How many pairs on duty can be chosen from *8* pupils?
4. How many handshakes take place among *12* people?

5. Find the number of diagonals of an octagon.